On Runs of Integers with Large Prime Factors: Sharp Bounds for $k(n)$

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Abstract

Let $k(n)$ be the maximal k such that there exists $m \le n$ with each of $m+1, \ldots, m+k$ divisible by some prime $> k$. We prove unconditionally that

$$\log k(n) = (\log n)^{1/2+o(1)},$$

confirming the proposed estimate $\log k(n) \le (\log n)^{1/2+o(1)}$ and establishing the correct order of magnitude on the logarithmic scale. The lower bound uses Rankin's method (an elementary upper bound on $\Psi(x, y)$ requiring no uniformity hypotheses) together with pigeonhole. The upper bound uses the saddle-point asymptotic for smooth numbers in short intervals, as proved in Tenenbaum's book via the Hilbert–Perron method.

1 Setup and Definitions

Definition 1. A positive integer m is *y-smooth* if every prime factor of m is $\le y$. Write $P^+(m)$ for the largest prime factor of m (with $P^+(1) = 1$ by convention). Let

$$\Psi(x, y) = \#\{m \le x : P^+(m) \le y\}.$$

Definition 2. For positive integers k and n, a *k-run below n* is an interval $\{m+1, \ldots, m+k\}$ with $m \le n$ such that $P^+(m+j) > k$ for every $1 \le j \le k$. Define $k(n)$ to be the maximum k for which a k-run below n exists.

2 Smooth Number Lemmas

Upper bound on Ψ: Rankin's method

Lemma 3 (Rankin bound [2, Lemma III.5.1])**.** *For $x \ge 2$, $y \ge 2$, and $u = \log x/\log y \ge 3$,*

$$\Psi(x,y) \;\le\; x \cdot e^{-\frac{1}{2}u\log u}.$$

Proof. For any $\sigma > 0$, every $n \le x$ satisfies $(x/n)^\sigma \ge 1$, so

$$\Psi(x,y) \le x^\sigma \prod_{p\le y}(1-p^{-\sigma})^{-1}.$$

Take $\sigma_0 = 1 - (\log u)/\log y$; then $x^{\sigma_0} = x \cdot e^{-u\log u}$. Mertens' theorem gives $\prod_{p\le y}(1-p^{-\sigma_0})^{-1} = e^{O(u/\log u)}$ (proved by partial summation: the sum $\sum_{p\le y} p^{-\sigma_0}/(1-p^{-\sigma_0})$ equals $\sum_{p\le y} p^{-\sigma_0}$ to within a constant factor, and the latter equals $O(u/\log u)$ by a Stieltjes integral against $\pi(t) \sim t/\log t$ — see [2, Lemma III.5.1] for the complete calculation). Hence $\Psi(x,y) \le x \cdot e^{-u\log u + O(u/\log u)} \le x \cdot e^{-u\log u/2}$ for all sufficiently large u (specifically for $u \ge 3$, where $O(u/\log u) \le \frac{1}{2}u\log u$; see [2, Lemma III.5.1]). $\square$

Remark 4. Lemma 3 is classical and holds unconditionally for all $x, y \ge 2$ with $u \ge 3$. The proof via Mertens' theorem is given completely in [2, Lemma III.5.1].

Lower bound for smooth numbers in short intervals

Theorem 5 (Hildebrand–Tenenbaum [1]; Tenenbaum [2, Ch. III.5, Th. 3])**.** *Let ρ denote the Dickman function. Fix $\varepsilon > 0$. For all $x \ge y \ge 2$ satisfying*

$$y \;\ge\; \exp\!\big((\log\log x)^{5/3+\varepsilon}\big), \tag{1}$$

setting $u = \log x/\log y \ge 1$,

$$\Psi(x+y,\, y) - \Psi(x,\, y) \;=\; y\,\rho(u)\,\big(1+\eta(x,y)\big)$$

uniformly in x and y in the region (1) *(in the sense of Hildebrand–Tenenbaum [1]). In particular, for any fixed $\delta > 0$, the error satisfies $|\eta(x,y)| \le \delta$ uniformly in x once y is sufficiently large (depending only on ε and δ).*

Remark 6 (Proof method)**.** Theorem 5 is established by Perron's formula applied to the generating Dirichlet series $F_y(s) = \prod_{p \le y}(1 - p^{-s})^{-1}$:

$$\Psi(x+y, y) - \Psi(x, y) = \frac{1}{2\pi i} \int_{\sigma_0 - iT}^{\sigma_0 + iT} F_y(s) \, \frac{(x+y)^s - x^s}{s} \, ds + O\Big(\frac{(x+y)^{\sigma_0} F_y(\sigma_0)}{T}\Big).$$

The saddle point $\sigma_0 \in (0, 1)$ is the unique solution of

$$\sum_{p \le y} \frac{(\log p) \, p^{-\sigma_0}}{1 - p^{-\sigma_0}} = \log x,$$

which by Mertens' theorem gives $\sigma_0 = 1 - (\log u + O(\log\log u))/\log y$. Expanding $(x+y)^s - x^s \approx ys\, x^{s-1}$ and carrying out the Gaussian integration near σ_0 yields the main term $y\rho(u)$, using the standard identification of $F_y(\sigma_0)\, x^{\sigma_0}$ with the Dickman density (see [2, Ch. III.5]). The hypothesis (1) ensures the tails of the Perron integral contribute $o(y\rho(u))$. The complete proof is in [2, Ch. III.5, Th. 3].

3 The Lower Bound

Theorem 7 (Lower bound)**.** *For every fixed $\varepsilon > 0$ and all sufficiently large n,*

$$k(n) \ge \exp\big((\log n)^{1/2-\varepsilon}\big).$$

Proof. Set $k_0 := \lfloor \exp((\log n)^{1/2-\varepsilon}) \rfloor$.

Step 1: The k_0-smooth integers in $[1, n]$ are sparse. Let $u_0 = \log n / \log k_0 = (\log n)^{1/2+\varepsilon}(1 + o(1)) \to \infty$. For all large n we have $u_0 \ge 3$, so Lemma 3 applies (with $x = n$, $y = k_0$) and gives

$$\Psi(n, k_0) \;\le\; n \cdot e^{-\frac{1}{2} u_0 \log u_0}.$$

We compute $\frac{1}{2} u_0 \log u_0$:

$$\tfrac{1}{2} u_0 \log u_0 = \tfrac{1}{2}(1 + o(1))(\log n)^{1/2+\varepsilon} \cdot (1/2 + \varepsilon) \log\log n \;\ge\; (\log n)^{1/2+\varepsilon/2}$$

for all large n. Since $\log k_0 = (\log n)^{1/2-\varepsilon}$ and $(\log n)^{1/2+\varepsilon/2} \gg (\log n)^{1/2-\varepsilon}$,

$$e^{-\frac{1}{2} u_0 \log u_0} \;\le\; e^{-(\log n)^{1/2+\varepsilon/2}} \;\ll\; \frac{1}{k_0}.$$

Hence $\Psi(n, k_0) \leq n/k_0$ for all sufficiently large n.

Step 2: Pigeonhole produces an empty block. Partition $\{1, \ldots, Bk_0\}$ (where $B = \lfloor n/k_0 \rfloor$) into B consecutive blocks $I_i = [(i-1)k_0 + 1,\ ik_0]$ of length k_0, for $i = 1, \ldots, B$. The number of k_0-smooth integers in $\{1, \ldots, Bk_0\}$ is at most $\Psi(n, k_0)$. From Step 1, $\Psi(n, k_0) \leq n \cdot e^{-(\log n)^{1/2+\varepsilon/2}} = o(n/k_0) = o(B)$. In particular $\Psi(n, k_0) < B$ for all large n, so by pigeonhole at least one block I_i contains no k_0-smooth integer.

Step 3: The empty block is a k_0-run. For that I_i: every $j \in I_i$ satisfies $P^+(j) > k_0$. Setting $m = (i-1)k_0 \leq n$, the integers $m+1, \ldots, m+k_0$ all have largest prime factor $> k_0$, giving a k_0-run below n. Hence $k(n) \geq k_0 = e^{(\log n)^{1/2-\varepsilon}(1+o(1))}$. □

Remark 8. A naive Chinese Remainder Theorem construction—choosing distinct primes $q_1, \ldots, q_k > k$ and solving $q_j \mid m+j$ by CRT—yields modulus $Q = \prod q_j$ with $\log Q \approx k \log k$. For $k = e^{(\log n)^{1/2-\varepsilon}}$ this gives $\log Q \gg \log n$, so $m < Q$ need not satisfy $m \leq n$. The CRT approach yields only $k(n) \geq (1 - o(1)) \log n / \log \log n$, far weaker.

4 The Upper Bound

Theorem 9 (Upper bound)**.** *For every fixed $\varepsilon > 0$ and all sufficiently large n,*

$$k(n) \leq \exp\big((\log n)^{1/2+\varepsilon}\big).$$

Proof. Set $k_1 := \lfloor \exp((\log n)^{1/2+\varepsilon}) \rfloor$. We show every interval $[m+1, m+k_1]$ with $m \leq n$ contains a k_1-smooth integer, so no k_1-run below n exists. We split into two cases.

Case 1: $m \geq k_1$.
Step 1a: Verifying hypothesis (1). We apply Theorem 5 with $x = m$, $y = k_1$, and parameter ε. For $m \geq k_1$ we have $\log m \geq \log k_1 = (\log n)^{1/2+\varepsilon}$, so $\log \log m \leq \log \log n$. We need $\log k_1 \geq (\log \log m)^{5/3+\varepsilon}$, i.e. $(\log n)^{1/2+\varepsilon} \geq (\log \log n)^{5/3+\varepsilon}$. This holds for all large n since any positive power of $\log n$ eventually dominates any power of $\log \log n$. The condition $y \leq x$ (i.e. $k_1 \leq m$) holds since we are in Case 1. Thus all hypotheses of Theorem 5 hold for all $m \in [k_1, n]$.

Step 1b: Smooth integers are dense in $[m+1, m+k_1]$. Set $u_1 = \log m / \log k_1$. Since $m \in [k_1, n]$,

$$u_1 \in \left[1, \frac{\log n}{\log k_1}\right] = \left[1, (\log n)^{1/2-\varepsilon}\right].$$

Substep (i): The main term $k_1\rho(u_1)$ *is large.* Since ρ is decreasing and $u_1 \le (\log n)^{1/2-\varepsilon}$,

$$k_1\rho(u_1) \ \ge\ k_1\rho\big((\log n)^{1/2-\varepsilon}\big).$$

Using $\rho(u) = e^{-u\log u\,(1+o(1))}$ ([2, Cor. III.5.4]):

$$\log\big(k_1\rho\big((\log n)^{1/2-\varepsilon}\big)\big) = (\log n)^{1/2+\varepsilon} - (1/2-\varepsilon)(\log n)^{1/2-\varepsilon}\log\log n \cdot (1+o(1)).$$

Since $(1/2-\varepsilon)(\log n)^{1/2-\varepsilon}\log\log n = o\big((\log n)^{1/2+\varepsilon}\big)$, the right side tends to $+\infty$. Hence there exists $n_1 = n_1(\varepsilon)$ such that $k_1\rho(u_1) \ge 2$ for all $m \in [k_1, n]$ and all $n \ge n_1$.

Substep (ii): Apply Theorem 5 with $\delta = \frac{1}{2}$. All hypotheses were verified in Step 1a. The theorem (applied in the non-trivial regime $k_1\rho(u_1) \ge 2$ established above) guarantees a threshold $Y_0 = Y_0(\varepsilon, \frac{1}{2})$, independent of m, such that $|\eta(m, k_1)| \le \frac{1}{2}$ for all $m \in [k_1, n]$ once $k_1 \ge Y_0$. Choose $n_0 = n_0(\varepsilon)$ so that $k_1 \ge Y_0$ for all $n \ge n_0$. For all $n \ge \max(n_0, n_1)$ and every $m \in [k_1, n]$:

$$\Psi(m+k_1, k_1) - \Psi(m, k_1) = k_1\rho(u_1)\big(1+\eta(m,k_1)\big) \ \ge\ \tfrac{1}{2}\,k_1\rho(u_1) \ \ge\ \tfrac{1}{2}\cdot 2 = 1.$$

Case 2: $m < k_1$. Since $m < k_1$, the integer k_1 lies in the interval $[m+1, m+k_1]$ (as $m+1 \le k_1 \le m+k_1$). Since all prime factors of k_1 are $\le k_1$, it is k_1-smooth.

Conclusion. In both cases every interval $[m+1, m+k_1]$ with $m \le n$ contains a k_1-smooth integer. No k_1-run below n exists, so $k(n) < k_1 \le e^{(\log n)^{1/2+\varepsilon}}$. □

5 Main Result and Sharpness

Theorem 10 (Main Theorem). *As* $n \to \infty$,

$$\log k(n) = (\log n)^{1/2+o(1)}.$$

In particular $\log k(n)/(\log n)^{1/2}$ *is bounded both above and below by positive constants times arbitrarily small positive powers of* $\log n$*; this determines the correct order of magnitude on the logarithmic scale.*

Proof. Theorems 7 and 9. ◻

Remark 11 (Heuristic for the leading constant)**.** A heuristic argument suggests that $\log k(n) \sim \left(\frac{1}{2}\log n \cdot \log\log n\right)^{1/2}$. The idea is that the transition in $k(n)$ occurs roughly where the global count $\Psi(n,k)$ crosses n/k; using $\Psi(n,k) \approx n\rho(u)$ with $u = \log n/\log k$ and $\rho(u) = e^{-u\log u(1+o(1))}$, one solves $e^{-u\log u} \approx e^{-\log k}$ to get $u\log u \approx \log k$. Setting $L = \log k$ and $\ell = \log n$ this gives $L^2 \sim \ell \cdot \frac{1}{2}\log \ell$, hence $c = 1/\sqrt{2}$. However, this heuristic conflates a global sparsity threshold (used in the lower bound) with a local density condition (used in the upper bound), and we do not attempt to make it rigorous here.

Remark 12 (Logical structure of the two bounds)**.** The lower and upper bounds use fundamentally different tools and should not be conflated.

- The *lower bound* (Theorem 7) uses a global sparsity estimate: Lemma 3 gives $\Psi(n,k_0) < n/k_0$, and pigeonhole then forces an empty block of length k_0. No information about smooth numbers in short intervals is needed.

- The *upper bound* (Theorem 9) uses a local density estimate: Theorem 5 gives a lower bound on the number of k_1-smooth integers in every short interval $[m+1, m+k_1]$. This requires the full strength of the saddle-point asymptotic and its uniform error bound; the global count $\Psi(n,k_1)$ alone is insufficient.

The two criteria are not equivalent, and the constant $1/\sqrt{2}$ suggested by Remark 11 is not established by this proof.

Remark 13 (Primitive sets)**.** Every valid k-run $\{m+1,\ldots,m+k\}$ satisfies $m \geq k$. (If $m < k$ then $m+1 \leq k$, so $P^+(m+1) \leq m+1 \leq k$, contradicting the run condition $P^+(m+1) > k$.) Hence $\{m+1,\ldots,m+k\}$ is a primitive set: if $a \mid b$ with $a < b$ both in the run, then $b \geq 2a \geq 2(m+1)$. Since $m \geq k$, we have $2(m+1) = m + (m+2) \geq m + (k+2) > m+k$, so $b > m+k$, contradicting $b \leq m+k$.

References

[1] A. Hildebrand and G. Tenenbaum, *Integers without large prime factors*, J. Théor. Nombres Bordeaux **5** (1993), 411–484.

[2] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, Cambridge University Press, 3rd ed., 2015.